Lab 3: Component Modelling & Architectural Pattern Selection

NAME:HEEMADHAWALA R
SRN:PES2UG24CS190
SECTION:D
SEM: 5

Architectural Justification

Student Project Portfolio & Showcase App

We chose Layered Architecture for the Student Project Portfolio & Showcase System.

Architectural Choice: The system is organized into three horizontal layers. The Presentation Layer (Showcase UI Component) serves Student Team, Panel Judge, and Public Voter interactions. The Business Layer holds four components: Project Submission Manager (FR-001, FR-002 – profile creation, artifact upload, GitHub URL validation), Judging & Rubric Component (FR-003 – digital rubric scoring), Voting & Anti-Sybil Component (FR-004 – one-vote-per-project enforcement), and Leaderboard Component (FR-005 – final ranking publication). The Data Layer (Database Component) persists all project, score, vote, and leaderboard records.

Reason 1 – Isolated, Auditable Business Rules: The showcase has several independent rule sets — repository/artifact validation, rubric scoring, one-vote-per-project anti-sybil checks, and deadline-gated leaderboard finalization. A layered structure keeps each rule set in its own Business-Layer component behind a stable interface, so the strict validation logic required by FR-001–FR-005 stays isolated from the UI and from each other, making the system easier to test and audit for a graded, GitHub-submitted lab deliverable.

Reason 2 – Fits an Event-Driven, Moderate-Scale Workload: Traffic is concentrated around a scheduled expo (submission window, judging period, a voting/leaderboard peak), not a constant high-scale service. A single layered deployment is far simpler for a small student team to build and operate than independently-deployed microservices, while still keeping the Data Layer as the single source of truth needed for NFR-002's no-data-loss and no-duplicate-vote guarantee.

Security Advantage: Because the Presentation Layer never talks to the Database Component directly, every vote must pass through the Voting & Anti-Sybil Component, which enforces verified-voter identity and the one-vote-per-project rule (FR-004) before anything is written to storage — centralizing anti-sybil and access-control checks in one auditable place rather than scattering them across the UI.

Performance Benefit: The Leaderboard Component can maintain a precomputed ranking updated incrementally from the Voting and Judging components' tallies rather than recalculating from raw vote/score records on every request, helping meet NFR-001's sub-300ms API latency target for the leaderboard under simulated peak voting load.